The heartwood is golden yellow to golden brown when freshly sawn, turning darker when exposed to light. Known for its durability as it is immune to the attack of termites and fungi.

The wood does not split or crack and is a carpenter friendly wood. It was the chief railway carriage and wagon wood in India. Ship building and bridge-building depends on teakwood. It is also used in making boats, toys, plywood, door frames and doors.

10.5 Latex

Rubber

Botanical name : *Hevea brasiliensis*

Family : Euphorbiaceae



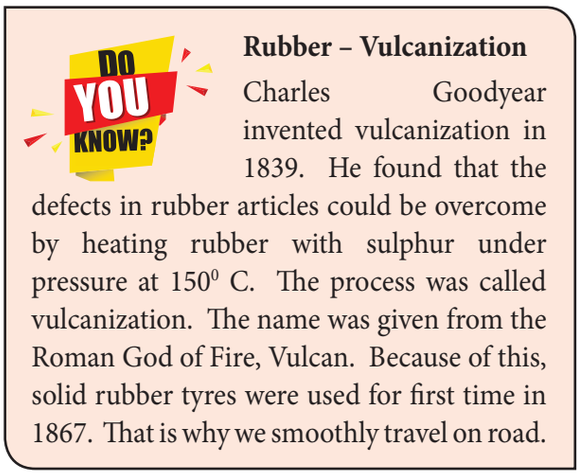
Figure 10.14: Rubber Tree

Origin and Area of cultivation: It is a native of Brazil and was introduced outside its native range during the colonial period and has become an important cash crop. Asia contributed 90% of the world production. Kerala is the largest producer in India followed by Tamil Nadu.

Uses

Tyre and other automobile parts manufacturing industries consume 70% of the rubber production. Rubber is used in manufacturing footwear, wire and cable insulations, rain-coats, household and hospital goods, shock absorbers, belts, sports goods, erasers, adhesives, and rubber-bands Hard rubber is used in the electrical and radio engineering industries Concentrated latex is used for making gloves, balloons and condoms. Foamed latex is used in the manufacture of cushions, pillows and life-belts.

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Rubber – Vulcanization

Charles Goodyear invented vulcanization in 1839. He found that the defects in rubber articles could be overcome by heating rubber with sulphur under pressure at 150^0 C. The process was called vulcanization. The name was given from the Roman God of Fire, Vulcan. Because of this, solid rubber tyres were used for first time in 1867. That is why we smoothly travel on road.

10.6 Pulp Wood



Figure 10.15 : Wood pulp

The term paper is derived from the word 'papyrus' a plant (*Cyperus papyrus*) that was used by Egyptians to make paper-like materials. Paper production is a Chinese invention. The Chinese discovered the paper that was prepared from the inner bark of paper mulberry in 105 A.D. For a long time, the art of paper making remained a monopoly of the Chinese until Arabs learned the technique and improved it around 750 A.D. Invention of printing increased the demand for paper.

Manufacture of Wood pulp: Wood is converted into pulp by mechanical, and chemical processes. Wood of Melia azadirachta, *Neolamarkia chinensis*, *Casuarina* spp, *Eucalyptus* spp are used for making paper pulp.



Purified dissolving pulp is used as a basic material in the manufacture of rayon or artificial silk, fabrics, transparent films (cellophane, cellulose acetate films), plastics. The viscose process of making rayon is the most common process.

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